

10-Mark Questions & Answers — Data Structures Module 3: Linked Lists

Q1. Explain Singly, Doubly, and Circular Linked Lists, Stack/Queue linked implementations, and Polynomial Addition in detail with complete C programs.

Comprehensive master answer covering:

1. Memory node layouts for SLL, DLL, and Circular Linked Lists.
2. Complete C code for Singly Linked List CRUD operations.
3. Doubly Linked List insertion and deletion with two-way pointer updates.
4. Linked Stack (Top at head) and Linked Queue (Front & Rear pointers) implementations.
5. Polynomial addition algorithm and complete C implementation.